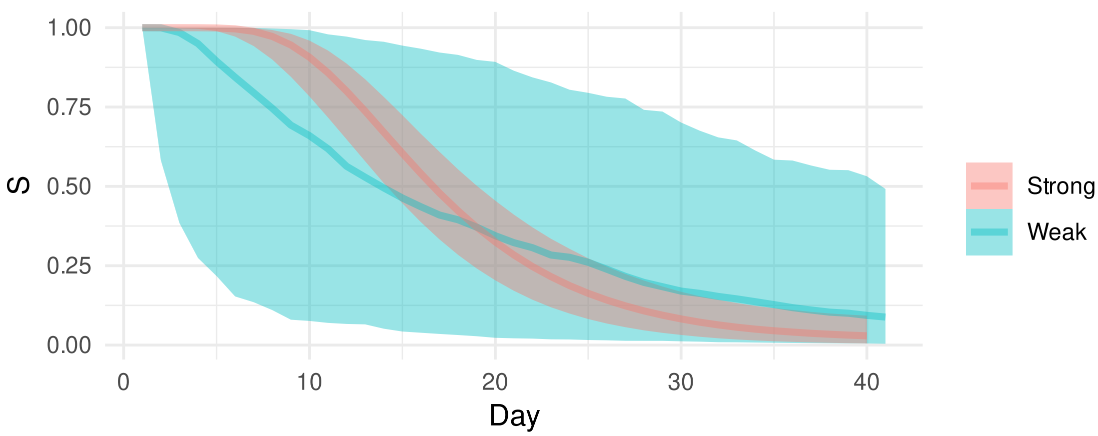
Supplementary Materials for "Estimating the duration of RT-PCR positivity for SARS-CoV-2 from doubly interval censored data with undetected infections"

by

Joshua Blake, Paul Birrell, A. Sarah Walker, Koen B. Pouwels, Thomas House, Brian D M Tom, Theodore Kypraios, and Daniela De Angelis



Web Figure 1: Prior predictive values of $S_{\boldsymbol{\theta}}$ for the two priors.

1 Web Appendix A: Details of survival function prior

The two priors we consider are depicted in Figure 1 They have varying degrees of informativeness regarding $S_{\boldsymbol{\theta}(t)}$.

1.1 Weakly informative prior

The first prior for $S_{\boldsymbol{\theta}}$ is weakly informative, centered on prior estimates. Specifically, we assume an independent prior distribution for each λ_t of Beta(0.1, 1.9). This distribution has mean 0.05, little information (standard deviation of 0.13), and a central 95% probability mass of 0.00–0.47. The mean is in line with previous estimates of the median duration (Cevik et al. 2020).

1.2 Strongly informative prior

The second is a strongly informative prior; it incorporates prior information from reliable estimates of λ_t for $t < 20$. We take a previous Bayesian analysis (Blake 2024) of data from The Assessment of Transmission and Contagiousness of COVID-19 in Contacts (ATACCC) study (Hakki et al. 2022), which tested individuals who had been exposed to infection daily up to a maximum of 20 days. This ATACCC-based analysis produces posterior estimates of λ_t with a posterior distribution with positive correlation between λ_t and $\lambda_{t'}$, especially for small $|t - t'|$. Furthermore, the uncertainty in the prior estimates for λ_t for $t \geq 20$ are underestimated because they are based on extrapolation of the ATACCC data under strong model assumptions.

We first approximate the previous posterior estimate of the hazard as logit $\boldsymbol{h} \sim \mathrm{N}(\boldsymbol{\mu}_A, \boldsymbol{\Sigma}_A)$ where $\boldsymbol{h}$ is the hazard, and $\boldsymbol{\mu}_A$ and $\boldsymbol{\Sigma}_A$ are the mean and covariance matrix estimated using samples of logit $\boldsymbol{h}$ from the previous study's posterior. Using a multivariate normal, as opposed to multiple univariate distribution for each h_t, preserves the correlation between the hazards. The approximation is very good (not shown).

Having approximated the estimate as a multivariate normal, we add additional uncertainty using a discrete Beta process. The discrete Beta process prior (Ibrahim et al. 2001, Sun 2006) generalizes the form of prior used in the weakly informative case by allowing the central estimate of the hazard to vary over time. It is:

$$\begin{aligned}
\operatorname{logit} \boldsymbol{h} &\sim \mathrm{N}(\boldsymbol{\mu}_A, \boldsymbol{\Sigma}_A) && (1)\\
\lambda_t &\sim \mathrm{Beta}(\alpha_t, \beta_t) \qquad t = 1, 2, \ldots && (2)\\
\alpha_t &= k_t h_t + \alpha_0 && (3)\\
\beta_t &= k_t(1 - h_t) + \beta_0, && (4)
\end{aligned}$$

where k_t, α_0, and β_0 are hyperparameters. An intuition for what this distribution represents derives from a conjugate model for λ_t with a beta prior and a binomial likelihood. If λ_t is given the prior distribution $\mathrm{Beta}(\alpha_0, \beta_0)$, and we then have k_t observations with $k_t h_t$ successes, then the posterior distribution for λ_t is $\mathrm{Beta}(\alpha_t, \beta_t)$ (as defined above). We use $\alpha_0 = 0.1$ and $\beta_0 = 1.9$ to match the weakly informative prior and the following form for k_t:

$$k_t = \begin{cases} \operatorname{expit}(-0.4 \times (t - 20)) & \text{for } t \leq 39 \\ 0 & t > 39. \end{cases} \tag{5}$$

The form and choice of constants reflects the subjective belief that h_t is a good estimate of λ_t for small t but increasingly unreliable; specifically, it is large at 0, when ATACCC is reliable, but becomes small for $t \geq 20$.

2 Web Appendix B: Derivations

2.1 Derivation of Equation (1)

Recall that n_k denotes the number of times that $\boldsymbol{\nu}_k$ appears in the episodes dataset (i.e. the observed data); n_u the latent number of undetected episodes; and $\boldsymbol{n} = [n_1, \ldots, n_{N_E}, n_\mathrm{u}]^T$. As we assume independence, the counts of each episode type are multinomially distributed:

$$\boldsymbol{n} \mid n_\mathrm{tot}, \boldsymbol{\theta} \sim \mathrm{Multinomial}(n_\mathrm{tot}, \boldsymbol{p}) \tag{6}$$

that is:

$$p(\boldsymbol{n} \mid n_\mathrm{tot}, \boldsymbol{\theta}) = \frac{n_\mathrm{tot}!}{n_\mathrm{u}! \prod_{k=1}^{N_E} n_k!} p_u^{n_\mathrm{u}} \prod_{k=1}^{N_E} p_k^{n_k}. \tag{7}$$

In the CIS data, each n_k ($k \neq u$) is observed as either 0 or 1. Define $\mathcal{D} = \{k : n_k = 1\}$, the set of detected episodes. Furthermore, note that the support of the multinomial distribution requires that $n_{\mathrm{u}} = n_{\mathrm{tot}} - n_{\mathrm{d}}$. Then Equation (7) simplifies to:

$$\begin{aligned} p(\boldsymbol{n} \mid n_{\mathrm{tot}}, \boldsymbol{\theta}) &= p(\boldsymbol{n}_{\mathrm{obs}} \mid n_{\mathrm{tot}}, \boldsymbol{\theta}) && (8) \\ &= \frac{n_{\mathrm{tot}}!}{(n_{\mathrm{tot}} - n_{\mathrm{d}})!} p_{\mathrm{u}}^{n_{\mathrm{tot}} - n_{\mathrm{d}}} \prod_{k \in \mathcal{D}} p_k. && (9) \end{aligned}$$

The relevant information from the CIS data is fully contained in the vector $\boldsymbol{n}_{\mathrm{obs}}$. Therefore, the posterior of interest is:

$$\begin{aligned} p(\boldsymbol{\theta} \mid \boldsymbol{n}_{\mathrm{obs}}) &\propto p(\boldsymbol{\theta}) p(\boldsymbol{n}_{\mathrm{obs}} \mid \boldsymbol{\theta}) && (10) \\ &= p(\boldsymbol{\theta}) \sum_{n_{\mathrm{tot}} = n_{\mathrm{d}}}^{\infty} p(n_{\mathrm{tot}}, \boldsymbol{n}_{\mathrm{obs}} \mid \boldsymbol{\theta}) && (11) \\ &= p(\boldsymbol{\theta}) \sum_{n_{\mathrm{tot}} = n_{\mathrm{d}}}^{\infty} p(n_{\mathrm{tot}} \mid \boldsymbol{\theta}) p(\boldsymbol{n}_{\mathrm{obs}} \mid n_{\mathrm{tot}}, \boldsymbol{\theta}) && (12) \\ &= p(\boldsymbol{\theta}) \sum_{n_{\mathrm{tot}} = n_{\mathrm{d}}}^{\infty} p(n_{\mathrm{tot}} \mid \boldsymbol{\theta}) \frac{n_{\mathrm{tot}}!}{(n_{\mathrm{tot}} - n_{\mathrm{d}})!} p_{\mathrm{u}}^{n_{\mathrm{tot}} - n_{\mathrm{d}}} \prod_{k \in \mathcal{D}} p_k \quad \text{by Equation (9)} && (13) \\ &= p(\boldsymbol{\theta}) \left(\prod_{k \in \mathcal{D}} p_k \right) \left(\sum_{n_{\mathrm{tot}} = n_{\mathrm{d}}}^{\infty} p(n_{\mathrm{tot}} \mid \boldsymbol{\theta}) \frac{n_{\mathrm{tot}}!}{(n_{\mathrm{tot}} - n_{\mathrm{d}})!} p_{\mathrm{u}}^{n_{\mathrm{tot}} - n_{\mathrm{d}}} \right). && (14) \end{aligned}$$

For convenience, define the summation term as:

$$\eta = \sum_{n_{\mathrm{tot}} = n_{\mathrm{d}}}^{\infty} p(n_{\mathrm{tot}} \mid \boldsymbol{\theta}) \frac{n_{\mathrm{tot}}!}{(n_{\mathrm{tot}} - n_{\mathrm{d}})!} p_{\mathrm{u}}^{n_{\mathrm{tot}} - n_{\mathrm{d}}}. \tag{15}$$

Next, we derive an analytical solution to η (defined in Equation (15)) assuming the prior $n_{\mathrm{tot}} \sim \mathrm{NegBin}(\mu, r)$, and that it is independent of $\boldsymbol{\theta}$, the parameters of the survival distribution. Therefore, $p(n_{\mathrm{tot}} \mid \boldsymbol{\theta}) = p(n_{\mathrm{tot}})$. This assumption makes η analytically tractable, allowing computationally feasible inference.

Putting a negative binomial prior on n_{tot} is equivalent to the following gamma-Poisson composite; its use simplifies the derivation.

$$\begin{aligned} n_{\mathrm{tot}} \mid \lambda &\sim \mathrm{Poisson}(\lambda) && (16) \\ \lambda &\sim \mathrm{Gamma}(a, b), && (17) \end{aligned}$$

where $b = r/\mu$ and $a = r$. Hence:

$$\eta = \int \sum_{n_{\mathrm{tot}}=n_{\mathrm{d}}}^{\infty} \frac{n_{\mathrm{tot}}!}{(n_{\mathrm{tot}}-n_{\mathrm{d}})!} p_{\mathrm{u}}^{n_{\mathrm{tot}}-n_{\mathrm{d}}} p(n_{\mathrm{tot}} \mid \lambda) p(\lambda) d\lambda \qquad \lambda \text{ explicit} \tag{18}$$

$$= \int \sum_{n_{\mathrm{tot}}=n_{\mathrm{d}}}^{\infty} \frac{n_{\mathrm{tot}}!}{(n_{\mathrm{tot}}-n_{\mathrm{d}})!} p_{\mathrm{u}}^{n_{\mathrm{tot}}-n_{\mathrm{d}}} \frac{\lambda^{n_{\mathrm{tot}}} e^{-\lambda}}{n_{\mathrm{tot}}!} p(\lambda) d\lambda \qquad n_{\mathrm{tot}} \sim \text{Poisson} \tag{19}$$

$$= \int \lambda^{n_{\mathrm{d}}} e^{-\lambda} p(\lambda) \sum_{n_{\mathrm{u}}=0}^{\infty} \frac{(p_{\mathrm{u}}\lambda)^{n_{\mathrm{u}}}}{n_{\mathrm{u}}!} d\lambda \qquad n_{\mathrm{u}} = n_{\mathrm{tot}} - n_{\mathrm{d}} \tag{20}$$

$$= \int \lambda^{n_{\mathrm{d}}} e^{-\lambda} p(\lambda) e^{\lambda p_{\mathrm{u}}} d\lambda \qquad \text{Maclaurin series of } e \tag{21}$$

$$= \int \lambda^{n_{\mathrm{d}}} e^{-\lambda(1-p_{\mathrm{u}})} \frac{b^a}{\Gamma(a)} \lambda^{a-1} e^{-b\lambda} d\lambda \qquad \lambda \sim \text{Gamma} \tag{22}$$

$$= \int \frac{b^a}{\Gamma(a)} \lambda^{a+n_{\mathrm{d}}-1} e^{-(b+1-p_{\mathrm{u}})\lambda} d\lambda \tag{23}$$

$$= \frac{b^a}{\Gamma(a)} \frac{\Gamma(a+n_{\mathrm{d}})}{(b+1-p_{\mathrm{u}})^{a+n_{\mathrm{d}}}} \qquad \text{Gamma pdf} \tag{24}$$

$$\propto (b+1-p_{\mathrm{u}})^{-(a+n_{\mathrm{d}})} \qquad \text{only } p_u \text{ depends on } \boldsymbol{\theta} \tag{25}$$

$$= (r/\mu+1-p_{\mathrm{u}})^{-(r+n_{\mathrm{d}})} \qquad \text{sub in } \mu \text{ and } r \tag{26}$$

$$\propto \{r+\mu(1-p_{\mathrm{u}})\}^{-(r+n_{\mathrm{d}})}. \tag{27}$$

Substituting into Equation (14) gives Equation (1).

2.2 Derivation of Equation (2)

To derive an expression for $p_{ik} = \Pr(O_j = \boldsymbol{\nu}_k \mid i_j = i_k, \boldsymbol{\theta})$, note that if $i_j = i_k$ then the event $O_j = \boldsymbol{\nu}_k$ occurs if and only if the episode starts in the interval $[l_k^{(b)}, r_k^{(b)}]$ and ends in the interval $[l_k^{(e)}, r_k^{(e)}]$. Omitting the conditioning on $\boldsymbol{\theta}$

and $i_j = i_k$, this gives:

$$\begin{aligned}
p_{ik} &= \Pr\left(l_k^{(b)} \leq B_j \leq r_k^{(b)}, l_k^{(e)} \leq E_j \leq r_k^{(e)}\right) && (28)\\
&= \Pr\left(l_k^{(e)} \leq E_j \leq r_k^{(e)} \mid l_k^{(b)} \leq B_j \leq r_k^{(b)}\right) \times \Pr\left(l_k^{(b)} \leq B_j \leq r_k^{(b)}\right) && (29)\\
&= \sum_{b=l_k^{(b)}}^{r_k^{(b)}} \Pr\left(l_k^{(e)} \leq E_j \leq r_k^{(e)} \mid B_j = b\right) \Pr\left(B_j = b\right) && (30)\\
&= \sum_{b=l_k^{(b)}}^{r_k^{(b)}} \Pr\left(l_k^{(e)} - b + 1 \leq D_j \leq r_k^{(e)} - b + 1\right) \Pr\left(B_j = b\right) \quad \text{by def of } D_j && (31)\\
&= \sum_{b=l_k^{(b)}}^{r_k^{(b)}} \left(S_{\boldsymbol{\theta}}(l_k^{(e)} - b + 1) - S_{\boldsymbol{\theta}}(r_k^{(e)} - b + 2)\right) \Pr\left(B_j = b\right) \quad \text{by def of } S_{\boldsymbol{\theta}} && (32)\\
&\propto \sum_{b=l_k^{(b)}}^{r_k^{(b)}} \left(S_{\boldsymbol{\theta}}(l_k^{(e)} - b + 1) - S_{\boldsymbol{\theta}}(r_k^{(e)} - b + 2)\right), && (33)
\end{aligned}$$

under the assumption of uniform prior probability of infection time. This is the standard form of the likelihood for doubly interval censored data without truncation (e.g. Sun 1995).

2.3 Derivation of Equation (3)

Recall that $p_u = \Pr(O_j = \varnothing \mid \boldsymbol{\theta})$, the probability that an arbitrary episode j is undetected. Therefore:

$$\begin{aligned}
1 - p_u &= 1 - \sum_{i=1}^{N_{\text{CIS}}} \Pr(O_j = \varnothing, i_j = i \mid \boldsymbol{\theta}) && (34)\\
&= 1 - \sum_{i=1}^{N_{\text{CIS}}} \Pr(O_j = \varnothing \mid i_j = i, \boldsymbol{\theta}) P(i_j = i \mid \boldsymbol{\theta}) && (35)\\
&= 1 - \frac{1}{N_{\text{CIS}}} \sum_{i=1}^{N_{\text{CIS}}} \Pr(O_j = \varnothing \mid i_j = i, \boldsymbol{\theta}) && (36)\\
&= \frac{1}{N_{\text{CIS}}} \sum_{i=1}^{N_{\text{CIS}}} (1 - \Pr(O_j = \varnothing \mid i_j = i, \boldsymbol{\theta})), && (37)
\end{aligned}$$

as we assume $P(i_j = i \mid \boldsymbol{\theta}) = 1/N_{\text{CIS}}$, that is all individuals are equally likely to be infected.

Let $p_{iu} = \Pr(O_j = \varnothing \mid i_j = i, \boldsymbol{\theta})$. An episode j in individual i_j is detected if and only if all the following conditions are met.

1. $t \in [B_j, E_j]$ for some $t \in \mathcal{T}_{i_j}$; i.e. there is at least one positive test for the episode.

2. $B_j > \min(\mathcal{T}_{i_j})$. For individuals enrolled during the period considered ($\min \mathcal{T}_{i_j} > 0$), this ensures that the beginning of the episode is lower bounded. For individuals enrolled prior to the period considered ($\min \mathcal{T}_{i_j} \leq 0$), this means that the episode was not detected prior to time 1.

3. $B_j \leq T_{i_j}$ where $T_{i_j} = \max\{t \in \mathcal{T}_{i_j} : t \leq T\}$ is the last time that i_j is tested in the period, meaning that the test is detected within the period.

4. $\exists t \in \mathcal{T}_{i_j}$ such that $t > E_j$, upper bounding the end of the episode. For episodes detected in the period we consider, a negligible number of episodes are excluded due to this criteria (<5% of detected episodes, themselves likely around 15% of all episodes). Therefore, we assume this occurs with probability 0.

First define $\tau_{\mathcal{T}_i}(t)$ as the time until the next test at or after time t in the schedule $\mathcal{T}_i$:

$$\tau_{\mathcal{T}_i}(t) = \min\{t' \in \mathcal{T}_i : t' \geq t\} - t. \tag{38}$$

The first condition can now be expressed as $e_j \geq b_j + \tau_{\mathcal{T}_{i_j}}(b_j)$. Equivalently, $d_j \geq \tau_{\mathcal{T}_{i_j}}(b_j) + 1$. Therefore, omitting the conditioning on $\boldsymbol{\theta}$ and $i_j = i$:

$$\begin{aligned}
1 - p_{iu} &= \Pr(D_j \geq \tau_{\mathcal{T}_i}(B_j) + 1, \min \mathcal{T}_i < B_j \leq T_i) && (39)\\
&= \sum_{b=\min \mathcal{T}_i+1}^{T_i} \Pr(D_j \geq \tau_{\mathcal{T}_i}(b) + 1 \mid B_j = b) \Pr(B_j = b) && (40)\\
&\propto \sum_{b=\min \mathcal{T}_i+1}^{T_i} S_{\boldsymbol{\theta}}(\tau_{\mathcal{T}_i}(b) + 1), && (41)
\end{aligned}$$

as we assume that all episode start times are independent with a uniform prior, meaning $\Pr(B_j = b)$ is a constant.

Substituting into Equation (37) gives Equation (3).

2.4 Derivation of Equation (6)

We now derive an expression for $p'_{ik} = \Pr(O'_j = \boldsymbol{\nu}'_k \mid i_j = i_k, \boldsymbol{\theta})$.

We proceed by first considering whether $E_j > r_k^{(e)}$ is the case and conditioning on $B_j = b$. Then, we combine the cases and remove the conditioning.

First, the case when $E_j \leq r_k^{(e)}$. In this case, the test at $r_k^{(e)} + 1$ is a true negative and the end of the episode is interval censored as in the case with no false negatives. The true negative occurs with probability 1, by the assumption of no false positives.

$$\Pr(O'_j = \boldsymbol{\nu}'_k, E_j \le r_k^{(e)} \mid B_j = b, i_j = i_k, p_{\text{sens}}, \boldsymbol{\theta}) \tag{42}$$
$$= \Pr(O'_j = \boldsymbol{\nu}'_k, l_k^{(e)} \le E_j \le r_k^{(e)} \mid B_j = b, i_j = i_k, p_{\text{sens}}, \boldsymbol{\theta}) \quad \text{the test at } l_k^{(e)} \text{ is positive} \tag{43}$$
$$= \Pr(O'_j = \boldsymbol{\nu}'_k \mid l_k^{(e)} \le E_j \le r_k^{(e)}, B_j = b, i_j = i_k, p_{\text{sens}}, \boldsymbol{\theta}) \tag{44}$$
$$\times \Pr(l_k^{(e)} \le E_j \le r_k^{(e)} \mid B_j = b, i_j = i_k, p_{\text{sens}}, \boldsymbol{\theta}) \tag{45}$$
$$= p_{\text{sens}}^{t_+}(1 - p_{\text{sens}})^{f_-} \left(S_{\boldsymbol{\theta}}(l_k^{(e)} - b + 1) - S_{\boldsymbol{\theta}}(r_k^{(e)} - b + 2) \right). \tag{46}$$

Second, the case when $E_j > r_k^{(e)}$. In this case, the test at $r_k^{(e)} + 1$ is a false negative, occurring with probability $(1 - p_{\text{sens}})$. To avoid having to consider tests after $r_k^{(e)}$, which could greatly complicate the likelihood, we model this case as the episode being right censored at $r_k^{(e)}$. Taking the same approach as before:

$$\Pr(O'_j = \boldsymbol{\nu}'_k, E_j > r_k^{(e)} \mid B_j = b, i_j = i_k, p_{\text{sens}}, \boldsymbol{\theta}) \tag{47}$$
$$= \Pr(O'_j = \boldsymbol{\nu}'_k \mid E_j > r_k^{(e)}, B_j = b, i_j = i_k, p_{\text{sens}}, \boldsymbol{\theta}) \tag{48}$$
$$\times \Pr(E_j > r_k^{(e)} \mid B_j = b, i_j = i_k, p_{\text{sens}}, \boldsymbol{\theta}) \tag{49}$$
$$= p_{\text{sens}}^{t_+}(1 - p_{\text{sens}})^{f_-}(1 - p_{\text{sens}}) S_{\boldsymbol{\theta}}(r_k^{(e)} - b + 2). \tag{50}$$

These expressions can now be used to derive p'_{ik}. First, augment the data with b, and split into the cases just discussed, omitting the conditioning on p_{sens}, $\boldsymbol{\theta}$, and $i_j = i_k$:

$$p'_{ik} = \Pr(O'_j = \boldsymbol{\nu}'_k) \tag{51}$$
$$= \sum_{b=l_k^{(b)}}^{r_k^{(b)}} \left(\Pr(O'_j = \boldsymbol{\nu}'_k, E_j \le r_k^{(e)} \mid B_j = b) + \Pr(O'_j = \boldsymbol{\nu}_k, E_j > r_k^{(e)} \mid B_j = b) \right) \Pr(B_j = b). \tag{52}$$

Now, substitute in Equations (46) and (50) and take out the common factor:

$$= p_{\text{sens}}^{t+}(1-p_{\text{sens}})^{f-} \tag{53}$$

$$\times \sum_{b=l_k^{(b)}}^{r_k^{(b)}} \left(S_{\boldsymbol{\theta}}(l_k^{(e)} - b + 1) - S_{\boldsymbol{\theta}}(r_k^{(e)} - b + 2) + (1-p_{\text{sens}}) S_{\boldsymbol{\theta}}(r_k^{(e)} - b + 2) \right) \tag{54}$$

$$\times \Pr(B_j = b \mid p_{\text{sens}}, \boldsymbol{\theta}) \tag{55}$$

$$= p_{\text{sens}}^{t+}(1-p_{\text{sens}})^{f-} \tag{56}$$

$$\times \sum_{b=l_k^{(b)}}^{r_k^{(b)}} \left(S_{\boldsymbol{\theta}}(l_k^{(e)} - b + 1) - p_{\text{sens}} S_{\boldsymbol{\theta}}(r_k^{(e)} - b + 2) \right) \tag{57}$$

$$\times \Pr(B_j = b \mid p_{\text{sens}}, \boldsymbol{\theta}). \tag{58}$$

We use a fixed p_{sens} (i.e. a point prior) and $\Pr(B_j = b \mid p_{\text{sens}}, \boldsymbol{\theta}) \propto 1$ giving:

$$p'_{ik} \propto \sum_{b=l_k^{(b)}}^{r_k^{(b)}} S_{\boldsymbol{\theta}}(l_k^{(e)} - b + 1) - p_{\text{sens}} S_{\boldsymbol{\theta}}(r_k^{(e)} - b + 2). \tag{59}$$

Estimating p_{sens} is not possible in the current framework (see Section 7).

2.5 Derivation of Equation (7)

The probability of one of the conditions that cause an episode to be missed (specified in Section 4.2) occurring, conditional on $B_j = b$ where $\min(\mathcal{T}_{i_j}) < b \leq T_{i_j}$ is:

$$\Pr\left(\tau_{\mathcal{T}_{i_j}}(b) + 1 \leq D_j \leq \tau^2_{\mathcal{T}_{i_j}}(b) \mid B_j = b, \boldsymbol{\theta}\right)(1-p_{\text{sens}}) \tag{60}$$

$$= \left(S_{\boldsymbol{\theta}}(\tau_{\mathcal{T}_{i_j}}(b) + 1) - S_{\boldsymbol{\theta}}(\tau^2_{\mathcal{T}_{i_j}}(b) + 1)\right)(1-p_{\text{sens}}). \tag{61}$$

Summing over b, in the same way as Equation (41), gives:

$$\zeta = (1-p_{\text{sens}})\frac{1}{T} \sum_{b=\min(\mathcal{T}_{i_j})+1}^{T_{i_j}} \left(S_{\boldsymbol{\theta}}(\tau_{\mathcal{T}_{i_j}}(b) + 1) - S_{\boldsymbol{\theta}}(\tau^2_{\mathcal{T}_{i_j}}(b) + 1)\right). \tag{62}$$

p'_{iu} is the probability of episode i being undetected, considering both the previous and new mechanisms. The previous and new mechanisms are mutually exclusive. Hence, p'_{iu} is the sum of these, $p'_{iu} = p_{iu} + \zeta$. As previously, $1 - p'_{iu}$

is the required quantity.

$$
\begin{aligned}
1 - p'_{iu} =& 1 - p_{iu} - \zeta && (63)\\
=& \frac{1}{T} \sum_{b=\min(\mathcal{T}_{i_j})+1}^{T_{i_j}} \left(p_{\text{sens}} S_{\boldsymbol{\theta}}(\tau_{\mathcal{T}_{i_j}}(b)+1) + (1-p_{\text{sens}}) S_{\boldsymbol{\theta}}(\tau^2_{\mathcal{T}_{i_j}}(b)+1) \right). && (64)
\end{aligned}
$$

3 Web Appendix C: Additional details of simulation study

3.1 Generation of data

The algorithm for generating the data is as follows.

1. Extract the test schedules for each individual who had at least one test during the period of interest.

2. Draw an episode start time, b_j for each individual uniformly at random between 2 July 2020 (100 days before the period where a detected episode would be included) and 6 December 2020 (the end of this period).

3. Draw a duration of episode for their episode, d_j, based on a combination of previous estimates (described below). Then calculate the end of their infection episode, $e_j = b_j + d_i - 1$.

4. Simulate the test results based on the test schedule, b_j, and e_j. A test on day t between b_j and e_j (inclusive) is positive with probability p_{sens}, where p_{sens} can vary with the time since infection, as defined below. All tests outside this interval are negative.

5. Discard episodes where there are no positive tests (i.e. undetected episodes) and then apply the inclusion criteria from Section 2. Denote by p the proportion of episodes that are retained.

6. Of these remaining episodes, sample $n_{\text{d}} = 4800$ to match the sample size of the true dataset. This is needed because in step 2 the entire cohort was infected, while in the real study only a (unknown) portion is infected.

7. For this final set of episodes, calculate $(l_j^{(b)}, r_j^{(b)}, l_j^{(e)}, r_j^{(e)})$ by taking the day after the last negative prior to any positives, the first positive, the last positive, and the day before the negative following the last positive respectively.

3.2 Distribution of duration

The simulation requires a distribution of the duration of detectability to use as ground truth. We modify the ATACCC-based duration estimate from Blake (2024, chapter 4) with an inflated tail to be consistent with the CIS. The tail inflation uses a simple survival analysis and the CIS data.

This analysis assumes the initiating event is known, and equal to the episode's detection time, $j_j^{(b)}$. It assumes the final event is interval censored between the time of the final positive test and the subsequent negative test, or right censored if a negative test has not yet been observed. A flexible, spline-based form is used for the baseline survival function (Royston 2014, Royston & Parmar 2002) with covariates introduced via proportional odds. By not accounting for either the undetected infections or the interval censoring of the initiating event, this analysis has competing biases which makes them hard to interpret (Office for National Statistics 2023*a*).

To form the duration distribution used in the simulation, we combine the two estimates. The pdf over the first 30 days is proportional to the ATACCC estimate, with the rest proportional to this CIS-based estimate. Denote by $f_A(t)$ the ATACCC-based distribution function and $f_C(t)$ that from the CIS-based estimates just derived. Then define $f'_S(t) = f_A(t)$ for $t \leq 30$ and $f'_S(t) = f_C(t)$ for $t > 30$. Then the distribution used in the simulation is the normalized version of this: $f_S(t) = f'_S(t) / \sum_i f'_S(i)$. Episode j's duration of detectability is then a draw from this distribution.

References

Blake, J. (2024), Estimating SARS-CoV-2 Transmission in England from Randomly Sampled Prevalence Surveys, PhD thesis, University of Cambridge, Cambridge, UK.

Cevik, M., Tate, M., Lloyd, O. & Maraolo, A. E. et al. (2020), 'SARS-CoV-2, SARS-CoV, and MERS-CoV viral load dynamics, duration of viral shedding, and infectiousness: A systematic review and meta-analysis', *The Lancet Microbe* **2**(1), e13–e22.

Hakki, S., Zhou, J., Jonnerby, J. & Singanayagam, A. et al. (2022), 'Onset and window of SARS-CoV-2 infectiousness and temporal correlation with symptom onset: A prospective, longitudinal, community cohort study', *The Lancet Respiratory Medicine* .

Ibrahim, J. G., Chen, M.-H. & Sinha, D. (2001), *Bayesian Survival Analysis*, Springer Series in Statistics, Springer New York, New York, NY.

Office for National Statistics (2022), 'Ethnic group, England and Wales', `https://www.ons.gov.uk/peoplepopulationandcommunity/culturalidentity/ethnicity/bulletins/ethnicgroupenglandandwales/census2021#cite-this-statistical-bulletin`.

Office for National Statistics (2023*a*), 'Coronavirus (COVID-19) Infection Survey: Methods and further information'.

Office for National Statistics (2023*b*), 'Rebasing of mid-year population estimates following Census 2021, England and Wales', `https://www.ons.gov.uk/peoplepopulationandcommunity/populationandmigration/populationestimates/bulletins/rebasingofmidyearpopulationestimatesfollowing/rebasingofmidyearpopulationestimatesfollowingcensus2021englandandwales#cite-this-statistical-bulletin`.

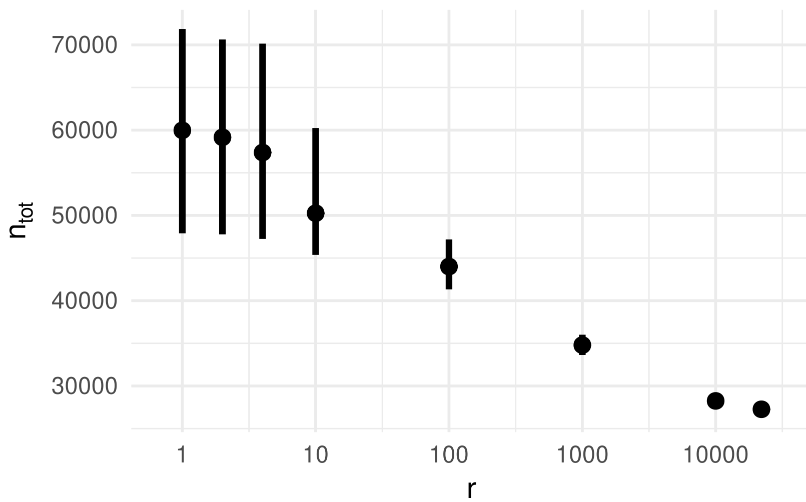
Royston, P. (2014), 'STPM: Stata module to fit flexible parametric models for survival-time data'.

Royston, P. & Parmar, M. K. B. (2002), 'Flexible parametric proportional-hazards and proportional-odds models for censored survival data, with application to prognostic modelling and estimation of treatment effects', *Statistics in Medicine* **21**(15), 2175–2197.

Sun, J. (1995), 'Empirical estimation of a distribution function with truncated and doubly interval-censored data and its application to AIDS studies', *Biometrics* **51**(3), 1096.

Sun, J. (2006), *The Statistical Analysis of Interval-Censored Failure Time Data*, Statistics for Biology and Health, Springer, New York.

4 Tables and figures



Web Figure 2: How the posterior estimate of n_{tot} changes with the value of r in the prior on n_{tot}.

	Cohort count	Cohort %	Population %
Age group			
Under 4	5320	1.6%	5.5%
5-9	13884	4.1%	5.9%
10-14	17709	5.2%	5.9%
15-19	14057	4.1%	5.6%
20-24	11565	3.4%	6.1%
25-29	15337	4.5%	6.6%
30-34	18999	5.6%	6.9%
35-39	21362	6.3%	6.7%
40-44	22372	6.6%	6.2%
45-49	24026	7.1%	6.5%
50-54	26544	7.8%	7.0%
55-59	28934	8.5%	6.8%
60-64	29529	8.7%	5.8%
65-69	29486	8.7%	5.0%
70-74	30203	8.9%	5.0%
75-79	17568	5.2%	3.6%
80-84	8866	2.6%	2.6%
85-89	3459	1.0%	1.5%
Over 90	1026	0.3%	0.9%
Sex			
Male	161212	47.4%	49.0%
Female	179034	52.6%	51.0%
Ethnicity			
White	314763	92.5%	81.7%
Asian	13449	4.0%	9.3%
Black	3134	0.9%	4.0%
Mixed	6048	1.8%	2.9%
Other	2852	0.8%	2.1%
Household size			
1	49716	14.6%	
2	139851	41.1%	
3	56129	16.5%	
4	63384	18.6%	
5+	31166	9.2%	
Country			
England	308539	90.7%	84.4%
Northern Ireland	7273	2.1%	2.8%
Scotland	14062	4.1%	8.1%
Wales	10372	3.0%	4.7%
Region			
North East	13906	4.1%	4.0%
North West	42725	12.6%	11.1%
Yorkshire and The Humber	29809	8.8%	8.2%
East Midlands	23015	6.8%	7.3%
West Midlands	27654	8.1%	8.9%
East	38032	11.2%	9.4%
London	59977	17.6%	13.3%
South East	45525	13.4%	13.8%
South West	27896	8.2%	8.5%

Web Table 1: Demographic information. Population % from 2020 mid-year population estimates (Office for National Statistics 2023*b*), except ethnicity which uses the England & Wales 2021 census (Office for National Statistics 2022). Population data source: Office for National Statistics licensed under the Open Government Licence v.3.0.